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Linear Algebra

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Autumn 2026

Definition

Let us choose a basis $B_{\mathcal{V}} = \{|a_i\rangle\}_{i=1}^N$ of a linear space $\mathcal{V}_N$, and express an arbitrary vector $|x\rangle$ in this basis: $|x\rangle = \xi_j |a_j\rangle$. We write

$$x = \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_N \end{pmatrix}$$

and say that the column vector x **represents** $|x\rangle$ in $B_{\mathcal{V}}$.

Remark

We can also have a linear homomorphism $\mathbf{A} \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$ act on the basis vectors in $B_{\mathcal{V}}$ to give vectors in the M -dimensional linear space $\mathcal{W}_M$: $|w_k\rangle = \mathbf{A}|a_k\rangle$. The latter can be written as a linear combination of basis vectors $B_{\mathcal{W}} = \{|b_j\rangle\}_{j=1}^M$ in $\mathcal{W}_M$:

$$|w_\ell\rangle = \alpha_{j\ell}|b_j\rangle.$$

The components can be arranged in a column as before to give a representation of the corresponding vectors:

$$\mathbf{w}_1 = \begin{pmatrix} \alpha_{11} \\ \alpha_{21} \\ \vdots \\ \alpha_{M1} \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} \alpha_{12} \\ \alpha_{22} \\ \vdots \\ \alpha_{M2} \end{pmatrix}, \quad \dots, \quad \mathbf{w}_N = \begin{pmatrix} \alpha_{1N} \\ \alpha_{2N} \\ \vdots \\ \alpha_{MN} \end{pmatrix}.$$

The operator itself is determined by the collection of all these vectors, i.e., by a matrix.

Definition

Let $\mathbf{A} \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$ and let $B_{\mathcal{V}} = \{|a_i\rangle\}_{i=1}^N, B_{\mathcal{W}} = \{|b_j\rangle\}_{j=1}^M$ be bases for $\mathcal{V}$ and $\mathcal{W}$, respectively. Define $\{\alpha_{ji}\}_{j=1,i=1}^{M,N}$ by $\mathbf{A}|a_i\rangle = \alpha_{ji}|b_j\rangle$. Then we write

$$\mathbf{A} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1N} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2N} \\ \vdots & \vdots & & \vdots \\ \alpha_{M1} & \alpha_{M2} & \cdots & \alpha_{MN} \end{pmatrix}$$

and call $\mathbf{A}$ the **matrix representing $\mathbf{A}$ in bases $B_{\mathcal{V}}$ and $B_{\mathcal{W}}$** .

Example 1.1

Find the matrix representation of the operator $\mathbf{T} \in \text{End}(\mathbb{R}^2)$ defined by

$$\mathbf{T} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}$$

in the canonical basis and in the basis B given by

$$B = \left\{ \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}.$$

Theorem 1.2

The two linear spaces $\text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$ and $\mathcal{M}^{M \times N}$ are isomorphic. An explicit isomorphism is established only when a basis is chosen for each linear space, in which case, an operator is identified with its matrix representation.

Notation

Let $\mathbf{A} \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$. Choose bases $B_{\mathcal{V}}$ for $\mathcal{V}$ and $B_{\mathcal{W}}$. We denote the matrix representing $\mathbf{A}$ in these bases by $\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}}(\mathbf{A})$, where

$$\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}} : \text{Hom}(\mathcal{V}_N, \mathcal{W}_M) \rightarrow \mathcal{M}^{M \times N}$$

is the basis-dependent linear isomorphism. When $\mathcal{V} = \mathcal{W}$, we leave out the subscripts and superscripts of $\mathbf{M}$, keeping in mind that all matrices are representations in a single basis.

Remark

Given the linear homomorphisms $\mathbf{A} : \mathcal{V}_N \rightarrow \mathcal{W}_M$ and $\mathbf{B} : \mathcal{W}_M \rightarrow \mathcal{U}_K$, we can form the composite $\mathbf{B} \circ \mathbf{A} : \mathcal{V}_N \rightarrow \mathcal{U}_K$. We can also choose bases $B_{\mathcal{V}} = \{|a_i\rangle\}_{i=1}^N$, $B_{\mathcal{W}} = \{|b_i\rangle\}_{i=1}^M$, $B_{\mathcal{U}} = \{|c_i\rangle\}_{i=1}^K$ for $\mathcal{V}$, $\mathcal{W}$, and $\mathcal{U}$, respectively. Then $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{B} \circ \mathbf{A}$ will be represented by an $M \times N$, a $K \times M$, and a $K \times N$ matrix, respectively, and we have

$$\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{U}}}(\mathbf{B} \circ \mathbf{A}) = \mathbf{M}_{B_{\mathcal{W}}}^{B_{\mathcal{U}}}(\mathbf{B})\mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}}(\mathbf{A}),$$

where on the right-hand side the product is defined as the usual product of matrices. If $\mathcal{V} = \mathcal{W} = \mathcal{U}$, we write the previous equation as

$$\mathbf{M}(\mathbf{B} \circ \mathbf{A}) = \mathbf{M}(\mathbf{B})\mathbf{M}(\mathbf{A}).$$

Notation

Matrices are determined entirely by their elements. For this reason a matrix A whose elements are $\alpha_{11}, \alpha_{12}, \dots$ is sometimes denoted by (α_{ij}) . Similarly, the elements of this matrix are denoted by $(A)_{ij}$. In the context of this notation, therefore, we can write

$$(A + B)_{ij} = (A)_{ij} + (B)_{ij} \implies (\alpha_{ij} + \beta_{ij}) = (\alpha_{ij}) + (\beta_{ij}),$$

$$(\gamma A)_{ij} = \gamma (A)_{ij} \implies \gamma (\alpha_{ij}) = (\gamma \alpha_{ij}),$$

$$(0)_{ij} = 0,$$

$$(1)_{ij} = \delta_{ij}.$$

Definition

The **transpose** of an $M \times N$ matrix A is an $N \times M$ matrix A^t obtained by interchanging the rows and columns of A :

$$(A^t)_{ij} = (A)_{ji}, \quad \text{or} \quad (\alpha_{ij})^t = (\alpha_{ji}).$$

Theorem 1.3

Let A and B be two matrices for which the operation of addition and/or multiplication are defined. Then

- i) $(A + B)^t = A^t + B^t;$
- ii) $(AB)^t = B^t A^t;$
- iii) $(A^t)^t = A.$

Theorem 1.4

Let $\mathbf{T} \in \text{End}(\mathcal{V}, \mathcal{W})$ and $B_{\mathcal{V}}$ and $B_{\mathcal{W}}$ be bases in $\mathcal{V}$ and $\mathcal{W}$. Let $\mathbf{T}^*, B_{\mathcal{V}}^*$, and $B_{\mathcal{W}}^*$ be duals to $\mathbf{T}$, $B_{\mathcal{V}}$, and $B_{\mathcal{W}}$, respectively. Let $\mathbf{T} \doteq \mathbf{M}_{B_{\mathcal{V}}}^{B_{\mathcal{W}}}(\mathbf{T})$ and $\mathbf{T}^* \doteq \mathbf{M}_{B_{\mathcal{W}}^*}^{B_{\mathcal{V}}^*}(\mathbf{T}^*)$. Then $\mathbf{T}^* = \mathbf{T}^t$.

Definition

A matrix S is **symmetric** if $S^t = S$. Similarly, a matrix A is **antisymmetric** if $A^t = -A$.

Remark

Any matrix A can be written as

$$A = \frac{1}{2} (A + A^t) + \frac{1}{2} (A - A^t),$$

where the first term is symmetric and the second is antisymmetric.

Definition

*A (real) matrix satisfying $A^t A = A A^t = 1$ is called **orthogonal**.*

Definition

Complex conjugation is an operations under which all elements of a matrix are complex conjugated. Denoting the complex conjugate of A by A^* , we have $(A^*)_{ij} = (A)_{ij}^*$, or $(\alpha_{ij})^* = (\alpha_{ij}^*)$.

Definition

The **hermitian conjugate** of a matrix A is the complex conjugation of the transpose, or, equivalently, the transpose of the complex conjugation. Thus, we have

$$A^\dagger = (A^t)^* = (A^*)^t,$$
$$(A^\dagger)_{ij} = (A)_{ji}^* \quad \text{or} \quad (\alpha_{ij})^\dagger = (\alpha_{ji}^*).$$

Definition

A **hermitian** matrix H satisfies $H^\dagger = H$, or, in terms of elements,

$$\eta_{ji}^* = \eta_{ij}.$$

A **unitary** matrix satisfies $UU^\dagger = U^\dagger U = 1$, or, in terms of elements,

$$\mu_{ik} \mu_{jk}^* = \mu_{ki}^* \mu_{kj} = \delta_{ij}.$$

Remark

- i) *The diagonal elements of a hermitian matrix are real.*
- ii) *The k th column of a hermitian matrix is the complex conjugate of its k th row, and vice versa.*
- iii) *A real hermitian matrix is symmetric.*
- iv) *The rows of an $N \times N$ unitary matrix, when considered as vectors in $\mathbb{C}^N$, form an orthonormal set under the usual inner product, as do the columns.*
- v) *A real unitary matrix is orthogonal.*

Definition

A matrix in which all of its off-diagonal elements are zero is called a **diagonal matrix**. A diagonal matrix whose diagonal elements are $\{\lambda_k\}_{k=1}^N$ is denoted by $\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N) \equiv \text{diag}(\lambda_k)_{k=1}^N$.

Example 1.5

Let $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ be a diagonal matrix, and $f(x)$ a function that has a Taylor series expansion around zero. Show that

$$f(D) = \text{diag}(f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)).$$

We list some familiar matrices in physics.

- i) A prototypical symmetric matrix is that of the moment of inertia encountered in mechanics. The ij th element of this matrix is defined as $I_{ij} = \int d^3x \rho(x_1, x_2, x_3) x_i x_j$, where x_i is the i th Cartesian coordinate of a point in the distribution of mass described by the volume density $\rho(x_1, x_2, x_3)$. It is clear that $I_{ij} = I_{ji}$, or $I = I^t$. The moment of inertia matrix can be represented as

$$I = \begin{pmatrix} I_{11} & I_{12} & I_{13} \\ I_{12} & I_{22} & I_{23} \\ I_{13} & I_{23} & I_{33} \end{pmatrix}$$

- ii) An example of an antisymmetric matrix is the electromagnetic field tensor given by

$$F = \begin{pmatrix} 0 & -B_3 & B_2 & E_1 \\ B_3 & 0 & -B_1 & E_2 \\ -B_2 & B_1 & 0 & E_3 \\ -E_1 & -E_2 & -E_3 & 0 \end{pmatrix}.$$

iii) Examples of hermitian matrices are the 2×2 **Pauli spin matrices**:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

iv) The most frequently encountered orthogonal matrices are rotations. One of such matrix represents the rotation of a 3-dimensional rigid body in terms of **Euler angles** and is used in mechanics. Attaching a coordinate system to the body, a general rotation can be decomposed into a rotation of angle φ about the z -axis, followed by a rotation of angle θ about the *new* x -axis, followed by a rotation of angle ψ about the *new* z -axis. We simply exhibit this matrix in terms of these angles:

$$\begin{pmatrix} \cos \psi \cos \varphi - \sin \psi \cos \theta \sin \varphi & -\cos \psi \sin \varphi - \sin \psi \cos \theta \cos \varphi & \sin \psi \sin \theta \\ \sin \psi \cos \varphi + \cos \psi \cos \theta \sin \varphi & -\sin \psi \sin \varphi + \cos \psi \cos \theta \cos \varphi & -\cos \psi \sin \theta \\ \sin \theta \sin \varphi & \sin \theta \cos \varphi & \cos \theta \end{pmatrix}.$$

Remark

The matrix representation of $\mathbf{A} \in \text{End}(\mathcal{V})$ is facilitated by choosing an orthonormal basis $B = \{|e_i\rangle\}_{i=1}^N$. The matrix elements of $\mathbf{A}$ can be found in such a basis by $(A)_{ij} = \alpha_{ij} = \langle e_i | \mathbf{A} | e_j \rangle$. We can also show that in an orthonormal basis, the i th component ξ_i of a vector is found by multiplying the vector by $\langle e_i |$. This expression for ξ_i allows us to write the expansion of $|x\rangle$ as

$$|x\rangle = \sum_{j=1}^N \underbrace{\langle e_j | x \rangle}_{\xi_j} |e_j\rangle = \sum_{j=1}^N |e_j\rangle \langle e_j | x \rangle \implies \mathbf{1} = \sum_{j=1}^N |e_j\rangle \langle e_j|,$$

which is the same as Theorem 3.27.

Remark

We may now calculate the matrix representing the hermitian conjugate of an operator $\mathbf{T}$. In an orthonormal basis, the elements of this matrix are given by $\tau_{ij} = \langle e_i | \mathbf{T} | e_j \rangle$. Taking the complex conjugate of this equation and using the definition of $\mathbf{T}^\dagger$ we get

$$\tau_{ij} = \langle e_i | \mathbf{T} | e_j \rangle^\dagger = \langle e_j | \mathbf{T} | e_i \rangle, \quad \text{or} \quad (\mathbf{T}^\dagger) = \tau_{ji}^*.$$

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Remark

Only in an orthonormal basis is the adjoint of an operator represented by the adjoint of the matrix representing that operator. In particular, a hermitian operator is represented by a hermitian matrix only if an orthonormal basis is used.

Example 1.6

Consider the matrix representation of the hermitian operator $\mathbf{H}$ in a general—not orthonormal—basis $B = \{|a_i\rangle\}_{i=1}^N$. The elements of the matrix corresponding to $\mathbf{H}$ are given by

$$\mathbf{H}|a_k\rangle = \sum_{j=1}^N \eta_{jk} |a_j\rangle, \quad \text{or} \quad \mathbf{H}|a_i\rangle = \sum_{j=1}^N \eta_{ji} |a_j\rangle.$$

By considering $\langle a_i | \mathbf{H} | a_k \rangle^*$ deduce that $\langle a_k | \mathbf{H} | a_i \rangle = \eta_{jk}^* \langle a_j | a_i \rangle$ and then, by multiplying the second equation by $\langle a_k |$ deduce that $\langle a_k | \mathbf{H} | a_i \rangle = \eta_{ji} \langle a_k | a_j \rangle$. Conclude that $\eta_{jk}^* \langle a_j | a_i \rangle = \eta_{ji} \langle a_k | a_j \rangle$. Show that if the $|a_i\rangle$'s are orthonormal, then we can conclude that $\eta_{ik}^* = \eta_{ki}$, as expected of a hermitian matrix.

Example 1.7

Let $\mathbf{U}$ be a unitary operator, i.e., $\mathbf{U}\mathbf{U}^\dagger = \mathbf{1}$ and let $\{|e_i\rangle\}$ be an orthonormal basis. In this case we have

$$\langle e_i | \mathbf{U}\mathbf{U}^\dagger | e_j \rangle = \langle e_i | \mathbf{1} | e_j \rangle = \delta_{ij}.$$

Insert the completeness relation $\mathbf{1} = |e_k\rangle\langle e_k|$ between $\mathbf{U}$ and $\mathbf{U}^\dagger$ to deduce that $\mu_{ik}\mu_{jk}^* = \delta_{ij}$. Similarly, doing the same calculation for $\mathbf{U}^\dagger\mathbf{U}$, deduce that $\mu_{ki}^*\mu_{kj} = \delta_{ij}$.

Remark

Given a basis $B = \{|b_i\rangle\}_{i=1}^N$, we can write an arbitrary vector $|a\rangle$ with components $\{\alpha_i\}_{i=1}^N$ in B as $|a\rangle = \alpha_i |b_i\rangle$. Now suppose that we change the basis to $B' = \{|b'_j\rangle\}_{j=1}^N$. We ask ourselves, how are the components of $|a\rangle$ in B' related to those in B ?

Definition

Let B, B' be bases for $\mathcal{V}_N$. Define the numbers $\{\rho_{ji}\}$ via $|b_i\rangle = \rho_{ji}|b_j\rangle$. The matrix

$$R = \begin{pmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1N} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2N} \\ \vdots & \vdots & & \vdots \\ \rho_{N1} & \rho_{N2} & \cdots & \rho_{NN} \end{pmatrix}$$

is called the **basis transformation matrix**.

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Corollary 1.8

The basis transformation matrix of any two bases B, B' is invertible.

Definition

Consider the equation $|b\rangle = \mathbf{A}|a\rangle$. This equation has a corresponding matrix equation $\mathbf{b} = \mathbf{A}\mathbf{a}$. The matrix $\mathbf{A}'$ defined by $\mathbf{b}' = \mathbf{A}'\mathbf{a}'$ satisfies following relations

$$\mathbf{R}^{-1}\mathbf{A}'\mathbf{R} = \mathbf{A}, \quad \text{or} \quad \mathbf{A}' = \mathbf{R}\mathbf{A}\mathbf{R}^{-1}.$$

This is called a **similarity transformation** on $\mathbf{A}$, and $\mathbf{A}'$ is said to be **similar** to $\mathbf{A}$.

Remark

Let $B = \{|e_i\rangle\}_{i=1}^N$ and $B' = \{|e'_i\rangle\}_{i=1}^N$ be two orthonormal bases of $\mathcal{V}_N$. Then the components of the transformation matrix R are given by the relation

$$\langle e'_j | e_i \rangle = \rho_{ji}.$$

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Theorem 1.9

The matrix that transforms one orthonormal basis into another is necessarily unitary.

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Example 1.10

Show that the similarity transform of a function of a matrix is the same function of the similarity transform of the matrix.

Definition

If A is representing $\mathbf{A}$ in some basis, then we set $\det \mathbf{A} = \det A$ (recall that the definition of determinant was independent of the basis).

Remark

Let $\mathbf{A}$ be a linear operator on $\mathcal{V}$ and let $\{|e_i\rangle\}$ be a basis of $\mathcal{V}$ in which $\mathbf{A}$ is represented by A . Then

$$\begin{aligned}
 \Delta_{\mathbf{A}}(|e_1\rangle, \dots, |e_N\rangle) &= \Delta(\mathbf{A}|e_1\rangle, \dots, |e_N\rangle) \\
 &= \Delta(\alpha_{i_1 1}|e_{i_1}\rangle, \dots, \alpha_{i_N N}|e_{i_N}\rangle) \\
 &= \alpha_{i_1 1} \dots \alpha_{i_N N} \Delta(|e_{i_1}\rangle, \dots, |e_{i_N}\rangle) \\
 &= \sum_{\pi \in S_N} \alpha_{\pi(1)1} \dots \alpha_{\pi(N)N} \Delta(|e_{\pi(1)}\rangle, \dots, |e_{\pi(N)}\rangle) \\
 &= \sum_{\pi \in S_N} \alpha_{\pi(1)1} \dots \alpha_{\pi(N)N} \epsilon_{\pi} \Delta(|e_1\rangle, \dots, |e_N\rangle)
 \end{aligned}$$

which by definition of determinant yields

$$\det \mathbf{A} = \det A = \sum_{\pi \in S_N} \alpha_{\pi(1)1} \dots \alpha_{\pi(N)N} \epsilon_{\pi} \equiv \sum_{\pi \in S_N} \epsilon_{\pi} \prod_{k=1}^N (A)_{\pi(k)k}.$$

Noting that $\sum_{\pi} = \sum_{\pi^{-1}}$ and $\epsilon_{\pi} = \epsilon_{\pi^{-1}}$ we have the following theorem.

Theorem 1.11

Let $\mathcal{V}$ be an N -dimensional linear space and let $\mathbf{A} \in \text{End}(\mathcal{V})$ and A its representation in any basis of $\mathcal{V}$. Then

$$\det \mathbf{A} = \det A = \sum_{\pi \in S_N} \epsilon_{\pi} \prod_{j=1}^N (A)_{\pi(j)j} = \epsilon_{i_1 i_2 \dots i_N} (A)_{i_1 1} \dots (A)_{i_N N}$$

$$\det \mathbf{A} = \det A^t = \sum_{\pi \in S_N} \epsilon_{\pi} \prod_{j=1}^N (A)_{j\pi(j)} = \epsilon_{i_1 i_2 \dots i_N} (A)_{1i_1} \dots (A)_{Ni_N}$$

where $\epsilon_{i_1 i_2 \dots i_N} = \epsilon_{\sigma(1)\sigma(2)\dots\sigma(p)} = \epsilon_{\sigma}$ with $i_k \equiv \sigma(k)$.

Theorem 1.12

Let A be a square matrix. Then

- i) $\det A$ is linear with respect to any row or column vector of A .*
- ii) If any two rows or two columns of A are interchanged, $\det A$ changes sign.*
- iii) Adding a multiple of one row [respectively, column] of A to another row [respectively, column] of A does not change $\det A$.*
- iv) $\det A = 0$ if and only if the rows [respectively, columns] are linearly dependent.*

Definition

Given any $N \times N$ matrix A , the determinant of the $(N - 1) \times (N - 1)$ matrix obtained from A by eliminating its i th row and j th column is called a **minor of order $N - 1$** and denoted by M_{ij} . The product $(-1)^{i+j} M_{ij}$ is called the **cofactor** of $(A)_{ij}$, and denoted by $(\text{cof } A)_{ij}$.

Theorem 1.13 (Laplace Expansion for the Determinant of a Matrix)

Given any $N \times N$ matrix A , the determinant $\det A$ of A is given by the following expression:

$$\det A = \sum_{j=1}^N (A)_{ji} (\text{cof } A)_{ji} = \sum_{j=1}^N (A)_{ij} (\text{cof } A)_{ij} \quad (\text{no sum on } i).$$

Example 1.14

Use the Laplace expansion to find the determinant of the matrix

$$A = \begin{pmatrix} 0 & i & -1 \\ 0 & 2 & -i \\ 1+i & a & b \end{pmatrix}$$

with $a, b \in \mathbb{C}$.

Theorem 1.15

The matrix A has an inverse if and only if $\det A \neq 0$. In this case we have

$$A^{-1} = \frac{(\text{cof } A)^t}{\det A}.$$

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Example 1.16

Compute the inverse of a general 2×2 matrix.

Definition

An **elementary row operation** on a matrix is one of the following:

- i) interchange of two rows of the matrix ,
- ii) multiplication of a row by a nonzero number, and
- iii) addition of a multiple of one row to another.

An **elementary column operation** is defined analogously.

Definition

A matrix is in **triangular** (or **row-echelon**) form if it satisfies the following three conditions:

- i) Any row consisting of only zeros is below any row that contains at least one nonzero element.
- ii) Going from left to right, the first nonzero entry of any row is to the left of the first nonzero entry of any lower row.
- iii) The first nonzero entry of each row is 1.

Theorem 1.17

For any invertible $n \times n$ matrix A , the $n \times 2n$ matrix $(A|1)$ can be transformed into the $n \times 2n$ matrix $(1|A^{-1})^\dagger$ by means of a finite number of elementary row operations.

† The matrix $(A|1)$ denotes the $n \times 2n$ matrix obtained by juxtaposing the $n \times n$ unit matrix to the right of A . It is easily shown that if A, B, C are $n \times n$ matrices, then $A(B|C) = (AB|AC)$.

Example 1.18

Evaluate the inverse of

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 2 & 1 & -1 \end{pmatrix}.$$

Verify your result.

Definition

*Given any $M \times N$ matrix A , an operator $\mathbf{T}_A \in \text{Hom}(\mathcal{V}_N, \mathcal{W}_M)$ can be associated with A , and one can construct the kernel and the range of $\mathbf{T}_A$. The rank of $\mathbf{T}_A$ is called the **rank** of A .*

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Theorem 1.19

The rank of a matrix is the dimension of the largest (square) submatrix whose determinant is not zero.

Definition

Let A be an $N \times N$ matrix. The mapping $\text{tr} : \mathcal{M}^{N \times N} \rightarrow \mathbb{C}$ given by $\text{tr } A = \alpha_{ij} \delta_{ij}$ is called the **trace** of A .

Theorem 1.20

The trace is a linear mapping. Furthermore,

$$\operatorname{tr}(AB) = \operatorname{tr}(BA) \quad \text{and} \quad \operatorname{tr} A^t = \operatorname{tr} A.$$

Theorem 1.21

To every operator $\mathbf{A} \in \text{End}(\mathcal{V})$ are associated two intrinsic numbers, $\det \mathbf{A}$ and $\text{tr } \mathbf{A}$, which are the determinant and the trace of the matrix representation of the operator in any basis of $\mathcal{V}$.

Corollary 1.22

The determinant of a diagonal matrix $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$ is given by the product of its diagonal elements:

$$\det D = \prod_{j=1}^N \lambda_j.$$

Theorem 1.23

For any two matrices $A, B \in \mathcal{M}^{N \times N}$, the mapping

$$g : \mathcal{M}^{N \times N} \times \mathcal{M}^{N \times N} \rightarrow \mathbb{C}$$

defined by $g(A, B) = \text{tr}(A^\dagger B)$ is a sesquilinear inner product.